

AYUSH M SINGH

Agra, Uttar Pradesh | +91 86 8199 6687 | ayushmsingh2004@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Computer Science undergraduate (Data Science specialization, CGPA 8.4/10) with three ML/Data Science internships and 12+ shipped end-to-end AI systems, including a 99.84%-accuracy deepfake-voice detection platform. Hackathon-winning AI/ML Lead experienced in directing AI coding assistants to accelerate model development, backend engineering, and deployment across the full stack. Proficient in Python, TensorFlow/PyTorch, SQL, and cloud deployment (GCP, Render, Netlify, Streamlit, Hugging Face Spaces, Cloudflare Pages).

EDUCATION

Vellore Institute of Technology (VIT), Vellore, India *Expected 2027*
B.Tech, Computer Science Engineering (Specialization: Data Science) | CGPA: 8.4/10 | No Standing Arrears
Higher Secondary (PCM): SSA Vidyapeeth, Agra — 79.6% (2023) | Secondary (Science): St. Paul's School, Agra — 94.6% (2021)

TECHNICAL SKILLS

Languages: Python, Java, R, C/C++, SQL
ML / Deep Learning: TensorFlow, Keras, PyTorch, Scikit-learn, LightGBM, XGBoost, CNN, YOLOv8, Grad-CAM, Transfer Learning
Generative AI & LLMs: Gemini, GPT, Claude, Prompt Engineering, RAG
AI-Assisted Development: Vibe-coding and AI-directed development using Claude, ChatGPT, Gemini, Hostinger Horizons & Antigravity IDE for rapid prototyping, code generation, debugging, and documentation across all projects
Full-Stack & APIs: FastAPI, React, Streamlit, REST APIs, WebSockets
Cloud & Deployment: GCP / Cloud Run, Docker, Render, Netlify, Hugging Face Spaces, Cloudflare Pages, Firebase
Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, MySQL, Power BI

PROFESSIONAL EXPERIENCE

Data Science Intern — Pinnacle Labs Pvt. Ltd. *Jun – Jul 2026*

- Built an AI-based Road Lane Detection system for ADAS/Autonomous Vehicles; evaluated classical CV and CNN segmentation approaches before fine-tuning YOLOv8 Nano via transfer learning on a custom annotated dataset.
- Built a Pascal VOC-to-YOLO annotation pipeline, evaluated the model via Precision/Recall/mAP, and deployed it as a live Gradio app on Hugging Face Spaces.

Data Science Intern — Skillified Mentor *May – Jul 2026*

- Built a Logistic Regression fraud-detection model on 284,807 credit card transactions (99.92% accuracy, 83% precision) and a Linear Regression temperature-prediction model on 701,548 sensor records.
- Built an ARIMA time-series forecasting model with seasonal decomposition; applied the full EDA-to-evaluation ML lifecycle across all three projects using Python, Pandas, and Scikit-learn.

AI/ML Intern — Edunet Foundation × AICTE (IBM SkillsBuild) *Jan – Feb 2026*

- Built and deployed an AI Meme & Poster Creator (Streamlit + Pillow) with a Pandas-based pipeline that cleaned 212,422 raw entries down to 10,196 curated captions (~95% reduction).
- Completed 8+ IBM SkillsBuild certifications in AI, Machine Learning, Deep Learning, NLP, and Computer Vision.

KEY PROJECTS

Audio Forensics — AI-Powered Voice Security Platform
Python, PyTorch, WavLM, LightGBM, FastAPI, Docker, Google Cloud Run

- Co-built a deepfake voice detection platform through 8+ model generations, fusing WavLM transformer embeddings with LFCC acoustic features via LightGBM to achieve 99.84% accuracy on 14,671 test samples.
- Built a FastAPI backend with REST + WebSocket real-time streaming inference, containerized with Docker and deployed on Google Cloud Run and Hugging Face Spaces.
- Built a multilingual dataset (9 languages, 90K clips) with GlotLID-based filtering to curb shortcut learning.

ConstructGuard AI — Building Defect Detection (Hackathon Winner)
Python, TensorFlow, Gemini 2.5 Flash, FastAPI, React, Grad-CAM

- Led a 3-member team as AI/ML Lead to build a 4-stage cascaded pipeline (Gemini validation → router → CNN specialists → Gemini report), achieving 76% accuracy across 9 defect categories; won a 36-hour national hackathon.
- Engineered the FastAPI backend (/predict, /health) integrating the router model, Moisture/Structural specialist classifiers, and Grad-CAM visualization with Gemini-generated reports; deployed the API to Hugging Face Spaces.

PoultryVision AI — Hierarchical Diagnostic Platform
Python, TensorFlow, CNN, Grad-CAM, Gemini, FastAPI, React

- Built a two-stage hierarchical CNN system (82% healthy/sick classification, 72% 4-disease classification) with Gemini-powered grounded veterinary recommendations, deployed full-stack on Netlify/Render.
- Combined original and Roboflow-augmented data via a custom cleaning pipeline; built the FastAPI + React (Vite) frontend.

ACHIEVEMENTS & CERTIFICATIONS

- Winner — Build & Beyond Hackathon + Designathon, VIT Vellore (Apr 2026): 1st place, Team Lead & AI/ML Lead for ConstructGuard AI.
- Published Author — IJPREMS: First-authored peer-reviewed literature review, "Attitudes to Language in Different Societies" (Nov 2024).
- Certifications:** IBM — AI, ML & Deep Learning, NLP & Computer Vision (8+ certificates); SmartBridge — Artificial Intelligence, in partnership with Google for Developers; Oracle Cloud — Certified AI Foundations Associate; Google — Prompt Design in Vertex AI, Gemini Multimodal RAG; Hackerrank — SQL(Basic)
- LeetCode:** Solved 300+ problems on Leetcode in Java & SQL; earned Top Interview 150, LeetCode 75, and Top SQL 50 badges.
- HackerRank:** Earned 5-Star Gold Badge in SQL and completed the SQL (Basic) Skill Certification.